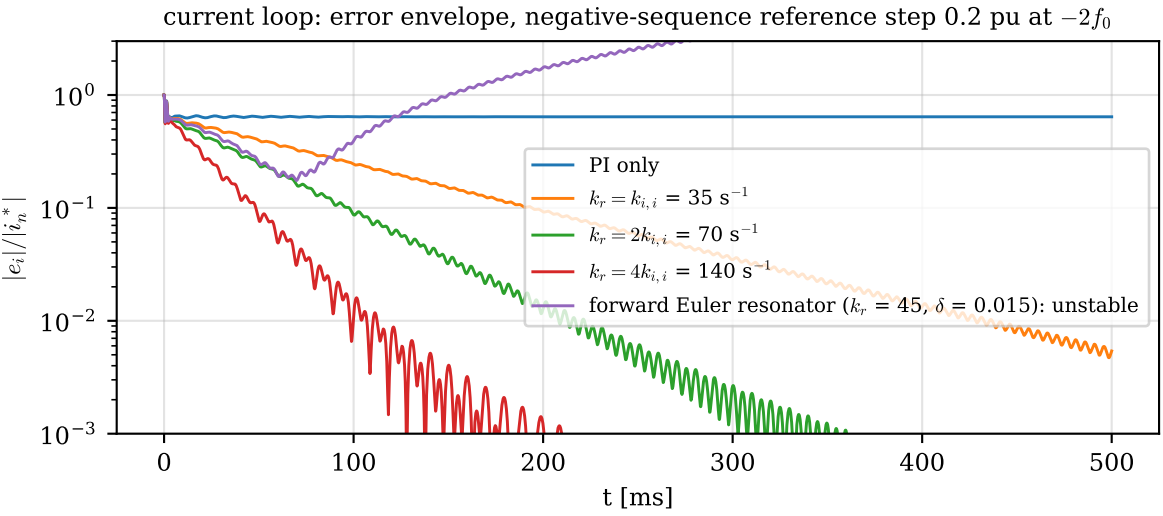

current loop: error envelope, negative-sequence reference step 0.2 pu at $-2f_0$
$|e_i|/|i_n^*|$
10^0
10^{-1}
10^{-2}
10^{-3}
0
100
200
300
400
500
t [ms]
PI only
$k_r = k_{i,i}$ = 35 s^{-1}
$k_r = 2k_{i,i}$ = 70 s^{-1}
$k_r = 4k_{i,i}$ = 140 s^{-1}
forward Euler resonator (k_r = 45, δ = 0.015): unstable